USRP X310 Data Recording Report

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Abstract

This report describes the data recording process using the USRP X310 device. This reciver is placed in a fixed position in São José dos Campos, SP, Brazil, in the GPS/GNSS Laboratory at Aeronautics Institute of Technology (ITA)[1]. The present report is updated as new recordings are made and new information is available.



Figure 1: Location of the GNSS antenna.

[1]https://github.com/ita-gnss-lab

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Part I
Recording overview

This part introduces an overview os the recordings presented in this report. It includes the methodology used to record the data, the calendar of the recorded days, the configuration of the USRP X310, and the processing of the recordings with GNSS-SDR.

1 Methodology

The recording is computed automatically by a Linux service which is driven by the following guidelines:

- Everyday, the program will start within the time rage [4PM, 6PM]
- It will sense whether the USRP is detectable via `uhd_find_devices --args=''addr=192.168.10.2''`.
- If not found, it will exit silently and will sense again in intervals of 15 min.
- Once found, two recordings are made, each with duration of 5 minutes (300 s). The first recording happens right after the USRP is detected, and the second recording is made 30 min after the first one. After that, no more recordings will be made for that day.
- The two recordings of a day use different signal pairs:
  - First recording: channel 0 (RX0) GPS L1 C/A (1575.42 MHz), 4 MS/s; channel 1 (RX1) GPS L5 (1176.45 MHz), 20 MS/s; both 25 dB of gain, signed 8-bit I/Q (2 bytes per complex sample);
  - Second recording: channel 0 (RX0) GPS L1 C/A (1575.42 MHz); channel 1 (RX1) GPS L2C (1227.60 MHz); both 4 MS/s, 25 dB of gain, signed 16-bit I/Q (4 bytes per complex sample).
- For each recording a SigMF archive file is created, with the interleaved I/Q samples + metadata, like number of channels (dual channel recording), frequencies, sampling rate, gain, sample type, etc. Each channel has its own configuration, so the metadata of each channel is reported separately.
- Each recording is processed by the GNSS-SDR software receiver right after it is made, and the outputs of every receiver stage are added under the recording in Section II (see Section 3). Galileo E1 shares GPS L1's 1575.42 MHz and Galileo E5a GPS L5's 1176.45 MHz, so the recordings also hold Galileo: the first recording is processed twice (GPS L1 C/A + L5, then Galileo E1 + E5a), and the second once, with GPS L1 C/A, L2C and Galileo E1 together.
- After 15 days of records from the last report. Rubem will send an automatic message to header of the Laboratory, Prof. Dr. Felix Antreich, via WhatsApp with the summarized information from all recorded metadata;

2 Recording calendar

Table 1 shows the days when recordings were made. The color of each day indicates how many recordings were made on that day: green means 2 recordings, blue means 1 recording, and no color means no recording was made.

October

M	T	W	T	F	S	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

Table 1: Recording calendar.

3 Validating the recordings with GNSS-SDR

A recording is only useful if a GNSS receiver can use it: find the satellites in it, track their signals, decode their navigation messages, and compute a position. To show that, every recording is processed with GNSS-SDR[2], an open-source software-defined GNSS receiver, and the output of each of its stages is added to the report of that recording (Section II). Figure 2 shows the data chain of GNSS-SDR. Each channel (one per satellite signal) runs its own acquisition, tracking and telemetry decoder; observables and PVT combine all of them.

[2] https://gnss-sdr.org, v0.0.20.

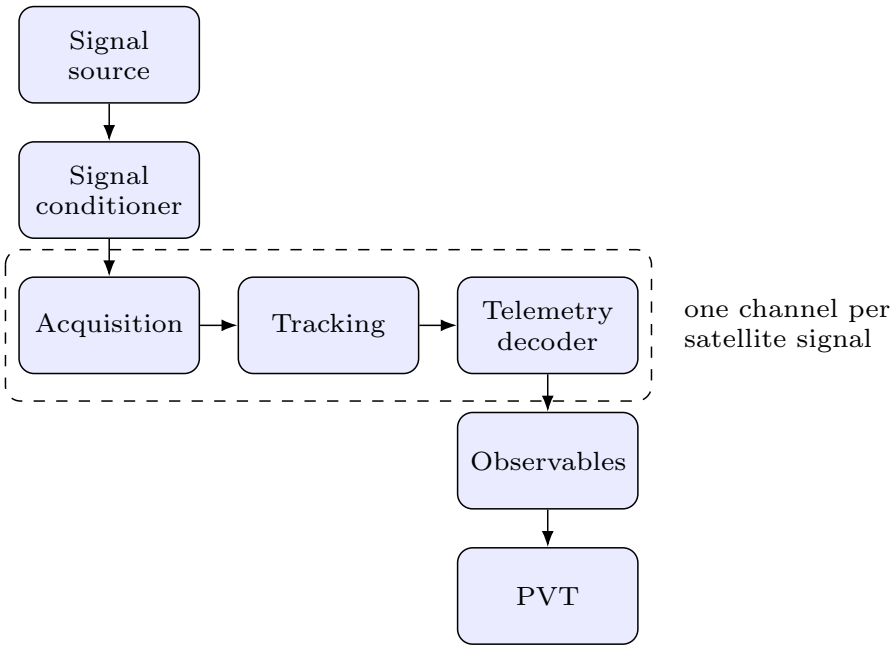


Figure 2: GNSS-SDR receiver chain.

Processing setup. Both channels of a recording are read by one `Multichannel_File_Signal_Source`, so the two bands advance in lock-step (with one independent source per band, one band runs ahead of the other and its measurements are discarded). That source reads both files at one rate: the 4 MS/s L1 C/A file of the L1/L5 recordings is upsampled 5 times (each sample repeated) to the 20 MS/s of L5 before GNSS-SDR. GNSS-SDR v0.0.20's position (RTKLIB, single point) is computed per run as follows:

- **First recording, GPS run** (GPS L1 C/A + L5): from L1 C/A only (broadcast ionosphere model), since its dual-frequency solver was unstable with the few L5-capable GPS satellites in view. The ionosphere-free GPS L1 C/A + L5 position is computed outside GNSS-SDR, from its observables and decoded ephemeris, with our own single-point positioning (`gnss_spp.py`).
- **First recording, Galileo run** (Galileo E1-B/C + E5a): GNSS-SDR's own dual-frequency, ionosphere-free (`Iono-Free-LC`) position. Every Galileo satellite broadcasts both E1 and E5a, so all satellites in view count, unlike GPS L5.
- **Second recording** (GPS L1 C/A + L2C + Galileo E1-B/C): GNSS-SDR's own single-frequency, multi-constellation position from GPS L1 C/A and Galileo E1 together (broadcast ionosphere model; RTKLIB estimates the GPS–Galileo clock offset). GNSS-SDR's solver ignores L2C, so the ionosphere-free GPS L1 C/A + L2C position again comes from `gnss_spp.py`.

The Galileo run's stages are reported after the GPS run's, with "(Galileo E1 + E5a)" in their headings; its signal source is the same recording, so that stage is not repeated.

3.1 Signal source

Reads the recorded I/Q samples (here, the SigMF data files) and delivers them as a stream, one per channel. GNSS-SDR has no output of its own here, so the recording itself is examined.

- **Power spectral density** per channel: the shape of the receiver passband over a flat thermal-noise floor. GNSS signals are 15–20 dB below the noise, so they are invisible here; a narrow peak would be interference.
- **I/Q histograms**: should be Gaussian, centred on zero, and far from the integer limits (no clipping).
- **Table**: format, rate and gain per channel, I/Q standard deviation and mean, percentage of clipped values, and samples lost during the recording (overflows, zero-filled).

3.2 Signal conditioner

Adapts the samples to what the channels process: converts the file format (signed 8- or 16-bit interleaved I/Q) to complex floating point (data type adapter), optionally filters (input filter) and resamples (resampler).

- **Table** per channel: file format, data type adapter, filter, resampling, and rate before and after.

3.3 Acquisition

Detects which satellites are present and gives a first, coarse estimate of each one's code delay and Doppler shift, by correlating the signal with the satellite's spreading code over a two-dimensional grid of delays and Doppler frequencies, and comparing the peak with a detection threshold (set by the probability of false alarm). With dual-frequency assistance, the second band (GPS L5 or L2C, Galileo E5a) is only searched for satellites already tracked on L1/E1, around their L1/E1 Doppler scaled to the second frequency. Galileo E1 uses a 4-ms code (BOC(1,1) modulation), so its correlation peak is narrower and has side peaks.

- **Delay–Doppler map** (surface plot) of the strongest detection per band: a single sharp peak well above the floor.
- **Table**: search attempts, detections and detected satellites, and the best test statistic relative to the threshold. Attempts on satellites that are not in view, or that do not broadcast that band (L5 is only on Block IIF and III satellites, L2C on IIR-M, IIF and III), are rejected.

3.4 Tracking

Follows each detected signal with a delay-lock loop (code) and a phase-lock loop (carrier), producing every 1–20 ms the prompt correlator output (which carries the navigation data bits), the carrier-to-noise density ratio C/N_0, the Doppler, the code, and carrier phases.

- C/N_0 **and Doppler versus time** per satellite and band: C/N_0 of roughly 35–50 dB-Hz for an open-sky antenna, and smooth Doppler curves following the satellite motion.
- **Prompt I/Q scatter** of the strongest satellite per band: with the carrier phase locked, the energy sits on the in-phase axis in two clusters (the ±1 data bits), or in a single cluster for the dataless pilot components that Galileo is tracked on (E1-C, E5a-Q); without phase lock it would be a ring. Each point is one prompt correlator output, the received samples $x[n]$ multiplied by the local code replica c aligned at the estimated code delay $\hat{\tau}$ and by the local carrier replica (estimated Doppler $\hat{f}_D$ and phase $\hat{\phi}$), summed over the N samples of one integration period (T_s: sampling period):

$$P = \sum_{n=1}^{N} x[n]\, c[n-\hat{\tau}]\, e^{-j(2\pi \hat{f}_D n T_s + \hat{\phi})} = I + jQ. \tag{1}$$

- **Table** per satellite: median C/N_0, percentage of time phase-locked, losses of lock and time tracked.

3.5 Telemetry decoder

Synchronizes to the navigation-message bits and frames, checks their parity/CRC (Cyclic Redundancy Check), decodes the ephemeris (orbit and satellite clock), and time-stamps every measurement with the satellite's transmit time (time of week). GPS L1 C/A carries the legacy LNAV message (6-s subframes); L2C and L5 carry CNAV (12-s and 6-s messages). Galileo E1-B carries I/NAV (2-s pages) and E5a-I F/NAV (10-s pages).

- **Timeline** per band and satellite of when a valid transmit time was available.
- **Table**: messages decoded, ephemeris obtained, and when the first valid time of week came.

3.6 Observables

Aligns all channels on common receiver epochs and forms, for each satellite and band, the pseudorange (from transmit and receive times), the accumulated carrier phase and the Doppler. These are the measurements a positioning algorithm uses; GNSS-SDR also writes them to a RINEX observation file.

- **Availability** of valid pseudoranges per band and satellite over the recording.

- **Second band minus L1/E1 pseudorange** per satellite: the same constant for every satellite (the receiver's inter-band hardware bias) plus the slowly varying, satellite-dependent ionospheric delay difference (metres). It proves that both bands measure the same satellites consistently.
- **Table** per satellite: common epochs, mean, standard deviation and drift of that difference.

3.7 PVT

Computes position, velocity and receiver clock (PVT) from the pseudoranges and the decoded ephemeris. The antenna position was surveyed: −23.209433°, −45.874875°, 596.42 m (WGS-84), so the errors below are true errors, not just scatter.

- **Horizontal error scatter and east/north/up error versus time** for GNSS-SDR's own solution of each run and, for the GPS runs, `gnss_spp.py`'s dual-frequency ionosphere-free solution. Single-frequency single-point positioning is expected to stay within a few metres horizontally and ∼10 m vertically; the iono-free combination removes the ionospheric bias but amplifies the code noise about three times.
- **Sky plot** of the GPS satellites used by `gnss_spp.py`.
- **Table**: mean error (accuracy), standard deviation (precision), 2-D and 3-D RMS, and 95% horizontal error.

Part II
Recording reports

4 Records on 08/10/2026

4.1 First record

Parameter	Channel 0	Channel 1
Start time	2026-10-08 22:26:37 UTC	
Frequency	1575.420 MHz (L1)	1176.450 MHz (L5)
Duration	300.0 s	
Sample rate	2.000 MS/s	
Sample type	signed complex 16-bit samples (I+Q), little-endian	
Gain	20.0 dB	20.0 dB
Antenna	RX1	RX2
File size	2400.0 MB	2400.0 MB
Samples written	600,000,000	600,000,000
Lost samples	0	0
Clock source	external	
Time source	internal	
Motherboard	USRP X310	
Daughterboard	TwinRX RX0	TwinRX RX1
Recorder	UHD Python API	

Table 2: Metadata for recording `x310_20261008T222630Z`, captured starting 2026-10-08T22:26:37.215640Z.

5 Records on 13/10/2026

5.1 First record

Parameter	**Channel 0**	**Channel 1**
Start time	2026-10-13 21:57:56 UTC	
Frequency	1575.420 MHz (L1)	1176.450 MHz (L5)
Duration	300.0 s	
Sample rate	2.000 MS/s	
Sample type	signed complex 16-bit samples (I+Q), little-endian	
Gain	20.0 dB	20.0 dB
Antenna	RX1	RX2
File size	2400.0 MB	2400.0 MB
Samples written	600,000,000	600,000,000
Lost samples	0	0
Clock source	external	
Time source	internal	
Motherboard	USRP X310	
Daughterboard	TwinRX RX0	TwinRX RX1
Recorder	UHD Python API	

Table 3: Metadata for recording `x310_20261013T215749Z`, captured starting 2026-10-13T21:57:56.746726Z.

5.2 Second record

Parameter	**Channel 0**	**Channel 1**
Start time	2026-10-13 22:08:41 UTC	
Frequency	1575.420 MHz (L1)	1176.450 MHz (L5)
Duration	300.0 s	
Sample rate	2.000 MS/s	
Sample type	signed complex 16-bit samples (I+Q), little-endian	
Gain	20.0 dB	20.0 dB
Antenna	RX1	RX2
File size	2400.0 MB	2400.0 MB
Samples written	600,000,000	600,000,000
Lost samples	0	0
Clock source	external	
Time source	internal	
Motherboard	USRP X310	
Daughterboard	TwinRX RX0	TwinRX RX1
Recorder	UHD Python API	

Table 4: Metadata for recording `x310_20261013T220835Z`, captured starting 2026-10-13T22:08:41.589855Z.

6 Records on 22/10/2026

6.1 First record

Parameter	Channel 0	Channel 1
Start time	2026-10-22 19:00:06 UTC	
Frequency	1575.420 MHz (L1)	1176.450 MHz (L5)
Duration	300.0 s	300.0 s
Sample rate	4.000 MS/s	20.000 MS/s
Sample type	signed complex 8-bit samples (I+Q), converted on the host from the X310's sc16	
Gain	25.0 dB	25.0 dB
Antenna	RX1	RX2
File size	2400.0 MB	12000.0 MB
Samples written	1,200,000,000	6,000,000,000
Lost samples	2912	14924
Clock source	external	
Time source	internal	
Motherboard	USRP X310	
Daughterboard	TwinRX RX0	TwinRX RX1
Recorder	UHD Python API	
GNSS-SDR configuration	`l1ca8bits4msps_l5_8bits20msps.conf`, `galileo_e1_8bits4msps_e5a_8bits20msps.conf`	

Table 5: Metadata for recording `x310_20261022T190000Z`, captured starting 2026-10-22T19:00:06.973177Z.

6.1.1 Signal source

Raw I/Q samples of each channel analysed within the window [10, 10.5] s after the start recording instant.

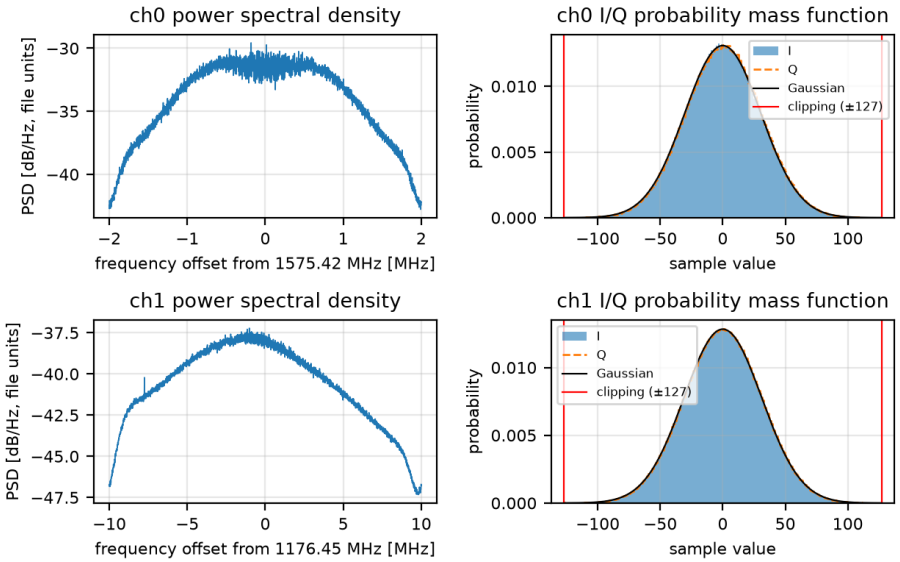


Figure 3: Signal source: power spectral density (the receiver passband over a flat thermal-noise floor; GNSS signals are below the noise, so only interference would stand out) and probability mass functions of the integer I/Q sample values (Gaussian noise, centred, not clipped).

Channel	**ch0**	**ch1**
Frequency [MHz]	1575.42	1176.45
Rate [MS/s]	4	20
Format	ci8	ci8
Bytes per complex sample	2 (1 for I, 1 for Q)	2 (1 for I, 1 for Q)
Full scale	±127	±127
Gain [dB]	25	25
Std I / Q	30.4 / 30.5	31.1 / 31.1
Mean I / Q	-0.26 / 0.20	0.26 / -0.07
Min / max I	-127 / 127	-127 / 127
Min / max Q	-127 / 127	-127 / 127
Headroom (full scale / std)	4.2	4.1
Clipped [%]	0.003	0.004
Lost samples	2,912	14,924

Table 6: Signal source: sample statistics per channel (headroom: how many standard deviations of the noise fit before clipping).

6.1.2 Signal conditioner

Channel	ch0 (GPS L1 C/A)	ch1 (GPS L5)
File format	ci8	ci8
Source item type	ibyte	ibyte
Data type adapter	Ibyte_To_Complex	Ibyte_To_Complex
Input filter	Pass_Through	Pass_Through
Resampling	×5 zero-order hold, offline (`upsample_iq.py`)	Pass_Through
Rate [MS/s]	4 → 20	20 → 20

Table 7: Signal conditioner: recorded format and rate to gnss-sdr's internal `gr_complex` stream; both bands are read in lockstep by one `Multichannel_File_Signal_Source`.

6.1.3 Acquisition

Two-dimensional search over code delay and Doppler (one channel per band dumped; the others behave alike).

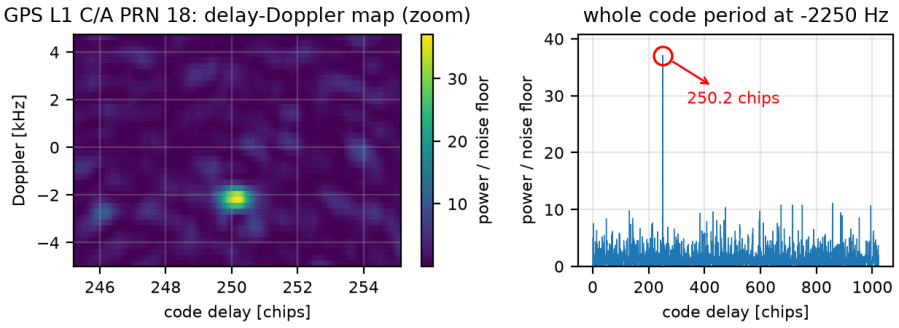


Figure 4: Acquisition, GPS L1 C/A: delay–Doppler search of PRN 18; one sharp peak (test statistic 50 vs threshold 47) at -2250 Hz and 250.2 chips.

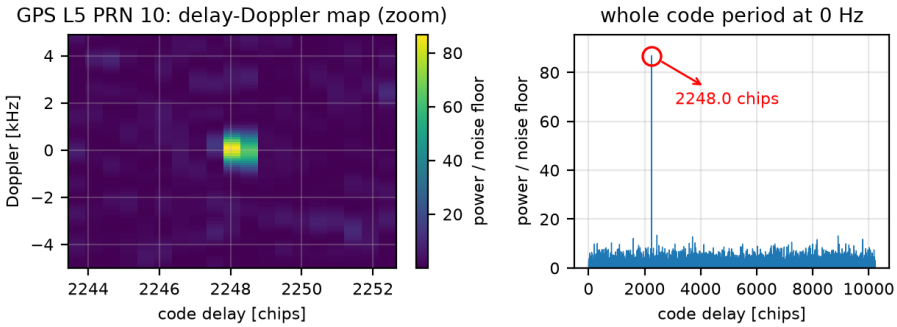


Figure 5: Acquisition, GPS L5: delay–Doppler search of PRN 10; one sharp peak (test statistic 122 vs threshold 49) at -1491 Hz and 2248.0 chips.

Band	Attempts	Detections	Detected PRNs	Best statistic / threshold
GPS L1 C/A	9	1	18	1.1
GPS L5	1	1	10	2.5

Table 8: Acquisition: search attempts of the dumped channel. Attempts on satellites that are not in view (or do not broadcast that signal, e.g. older GPS satellites on L5/L2C) are rejected below the threshold.

6.1.4 Tracking

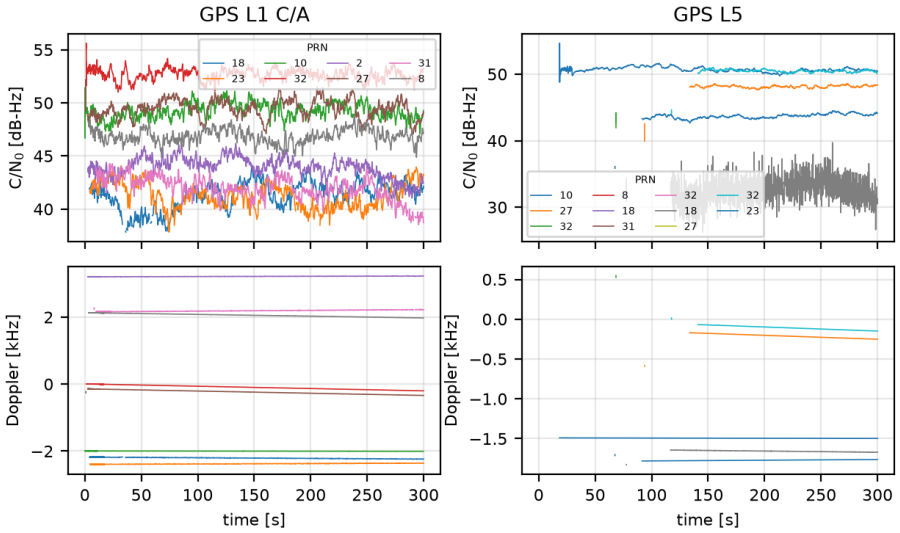


Figure 6: Tracking: carrier-to-noise density ratio and carrier Doppler per satellite, while phase-locked (smooth Doppler curves = loops following the real satellite motion).

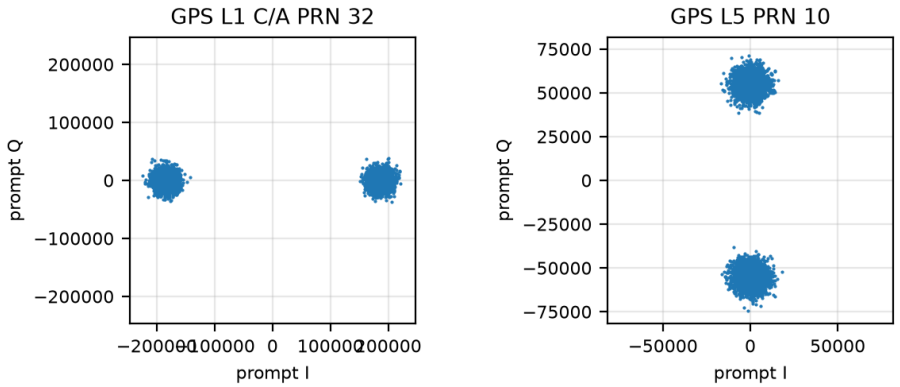


Figure 7: Tracking: prompt correlator outputs of the strongest satellite per band. Two tight clusters on one axis (the ±1 data bits; GNSS-SDR's L5 tracking puts the data component on the Q axis), or a single one for a dataless pilot (Galileo E1-C and E5a-Q, tracked here), mean the PLL is phase-locked; a ring would mean it is not. `Prompt_I`/`Prompt_Q` are 32-bit floats (sums of the samples over one integration period, in file units): they never clip, only overflowing beyond $\pm 3.4 \times 10^{38}$, far above the $N \cdot \text{full scale} \cdot \sqrt{2}$ a sum of N samples can reach.

Band	Ch.	PRN	Median C/N_0 [dB-Hz]	Phase-locked [%]	Losses of lock	Tracked [s]
GPS L1 C/A	0	18	41.7	93	0	298
GPS L1 C/A	1	23	41.9	93	0	297
GPS L1 C/A	2	10	49.2	100	0	300
GPS L1 C/A	3	32	52.9	100	0	299
GPS L1 C/A	4	2	43.9	95	0	299
GPS L1 C/A	5	27	49.1	94	0	299
GPS L1 C/A	6	31	42.7	94	0	292
GPS L1 C/A	7	8	47.0	94	0	298
GPS L5	8	10	50.7	100	0	282
GPS L5	10	27	48.0	37	2	206
GPS L5	10	32	29.0	0	2	16
GPS L5	11	8	28.4	0	2	16
GPS L5	11	18	33.5	0	2	16
GPS L5	12	31	29.0	0	2	16
GPS L5	12	32	28.7	0	2	16
GPS L5	13	18	32.8	60	1	206
GPS L5	13	27	28.8	0	1	16
GPS L5	14	32	50.3	36	1	198
GPS L5	15	23	43.7	63	0	232

Table 9: Tracking: per channel and satellite (phase-locked: carrier lock test > 0.8).

6.1.5 Telemetry decoder

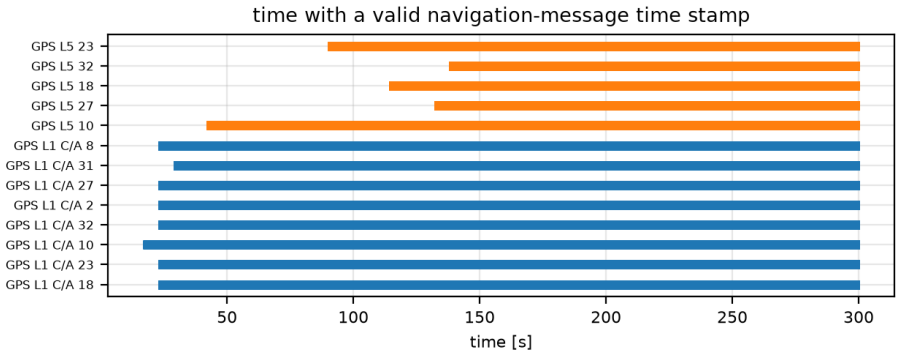


Figure 8: Telemetry decoder: per band and satellite, when the decoder delivered a valid transmit time (the first one comes after the first complete frame/message).

Band	PRN	Messages decoded	Ephemeris	First valid at [s]	TOW [s]
GPS L1 C/A	2	47	yes	23.1	414048.0
GPS L1 C/A	8	45	yes	23.1	414048.0
GPS L1 C/A	10	43	yes	17.1	414042.0
GPS L1 C/A	18	47	yes	23.1	414048.0
GPS L1 C/A	23	47	yes	23.1	414048.0
GPS L1 C/A	27	47	yes	23.1	414048.0
GPS L1 C/A	31	46	yes	29.1	414054.0
GPS L1 C/A	32	47	yes	23.1	414048.0
GPS L5	10	26	yes	42.2	414067.1
GPS L5	18	24	yes	114.4	414139.3
GPS L5	23	27	yes	90.0	414114.9
GPS L5	27	17	yes	132.3	414157.2
GPS L5	32	16	yes	138.2	414163.1

Table 10: Telemetry decoder: GPS L1 C/A LNAV subframes, GPS L5 CNAV messages decoded, ephemeris obtained, and the first valid time of week.

6.1.6 Observables

Pseudorange, carrier phase and Doppler of every channel at common receiver epochs.

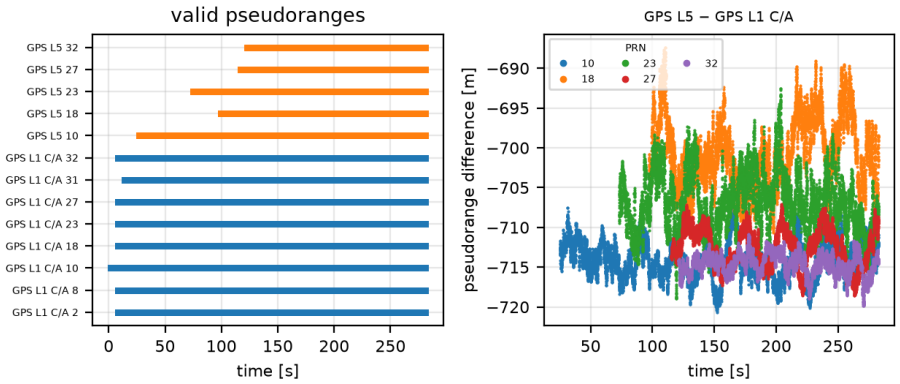


Figure 9: Observables: epochs with a valid pseudorange per band and satellite, and the difference between the two bands' pseudoranges of the same satellite (a constant receiver inter-band bias, the same for every satellite, plus the slowly varying ionospheric delay).

Difference	PRN	Common epochs	of run [%]	Mean [m]	Std [m]	Drift [m/s]
GPS L5 – GPS L1 C/A	10	12831	86	-714.11	2.26	0.0005
GPS L5 – GPS L1 C/A	18	9267	62	-700.79	4.40	0.0122
GPS L5 – GPS L1 C/A	23	10483	70	-706.92	3.69	-0.0064
GPS L5 – GPS L1 C/A	27	8373	56	-712.59	2.40	-0.0092
GPS L5 – GPS L1 C/A	32	8078	54	-714.63	1.36	-0.0046

Table 11: Observables: pseudorange difference between the two bands of each satellite at common epochs (similar means = shared receiver bias; std = code noise of both bands).

6.1.7 PVT

Position from the pseudoranges and the decoded ephemeris, compared with the surveyed antenna position (-23.209433°, -45.874875°, 596.42 m).

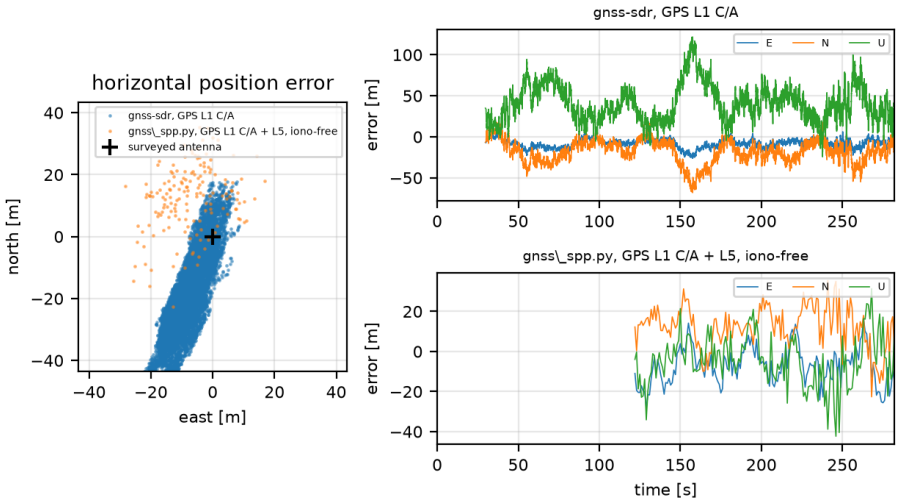


Figure 10: PVT: position error with respect to the surveyed antenna position. gnss-sdr: GNSS-SDR's own RTKLIB solution (single frequency, broadcast ionosphere model). gnss_spp.py: computed from gnss-sdr's observables and decoded GPS ephemeris, removing the ionospheric delay with the second band (pseudoranges carrier-smoothed over 100 s).

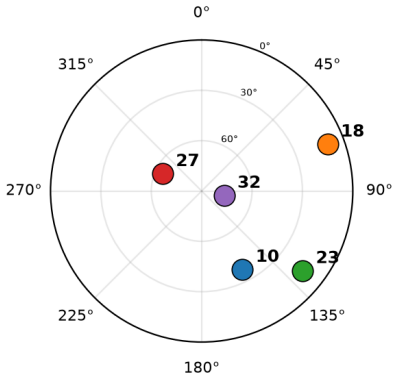


Figure 11: PVT: sky plot of the GPS satellites in the gnss_spp.py dual-frequency solution (azimuth, elevation) during the recording.

Solution	gnss-sdr, GPS L1 C/A	gnss_spp.py, GPS L1 C/A + L5, iono-free
Solutions	13135	161
Run with fix [%]	90	57
Mean E / N / U [m]	-7.6 / -18.0 / 40.8	-7.3 / 12.2 / -4.6
Std E / N / U [m]	5.5 / 14.2 / 22.4	9.3 / 10.4 / 12.8
RMS 2-D [m]	24.8	19.9
RMS 3-D [m]	52.8	24.1
95% horiz. [m]	46.5	29.6

Table 12: PVT: position error statistics with respect to the surveyed antenna (mean = accuracy, std = precision; % of the run's seconds with a position; gnss-sdr outputs several solutions per second, gnss_spp.py computes one per second).

6.1.8 Signal conditioner (Galileo E1 + E5a)

Channel	ch0 (Galileo E1)	ch1 (Galileo E5a)
File format	ci8	ci8
Source item type	ibyte	ibyte
Data type adapter	Ibyte_To_Complex	Ibyte_To_Complex
Input filter	Pass_Through	Pass_Through
Resampling	×5 zero-order hold, offline (`upsample_iq.py`)	Pass_Through
Rate [MS/s]	4 → 20	20 → 20

Table 13: Signal conditioner: recorded format and rate to gnss-sdr's internal `gr_complex` stream; both bands are read in lockstep by one `Multichannel_File_Signal_Source`.

6.1.9 Acquisition (Galileo E1 + E5a)

Two-dimensional search over code delay and Doppler (one channel per band dumped; the others behave alike).

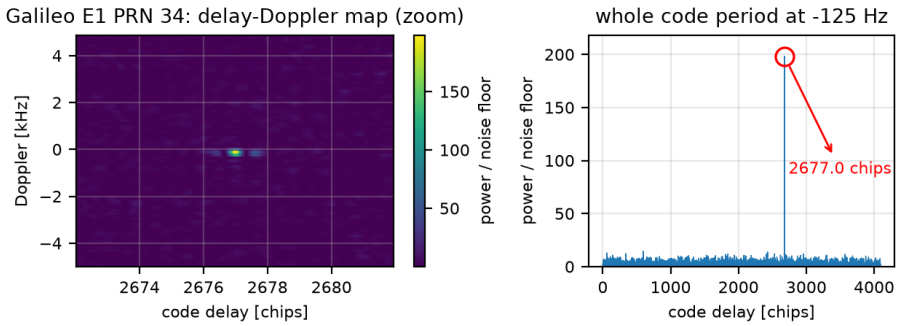


Figure 12: Acquisition, Galileo E1: delay–Doppler search of PRN 34; one sharp peak (test statistic 272 vs threshold 52) at -125 Hz and 2677.0 chips.

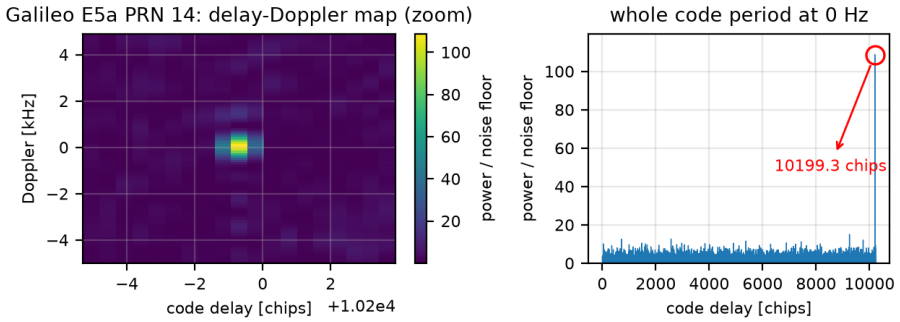


Figure 13: Acquisition, Galileo E5a: delay–Doppler search of PRN 14; one sharp peak (test statistic 149 vs threshold 49) at -1427 Hz and 10199.3 chips.

Band	Attempts	Detections	Detected PRNs	Best statistic / threshold
Galileo E1	5	1	34	5.3
Galileo E5a	2	2	14	3.0

Table 14: Acquisition: search attempts of the dumped channel. Attempts on satellites that are not in view (or do not broadcast that signal, e.g. older GPS satellites on L5/L2C) are rejected below the threshold.

6.1.10 Tracking (Galileo E1 + E5a)

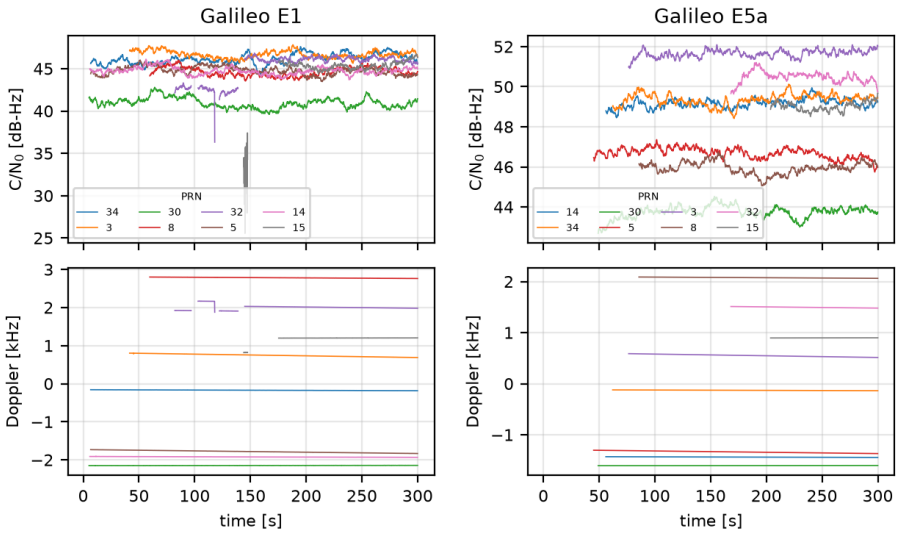


Figure 14: Tracking: carrier-to-noise density ratio and carrier Doppler per satellite, while phase-locked (smooth Doppler curves = loops following the real satellite motion).

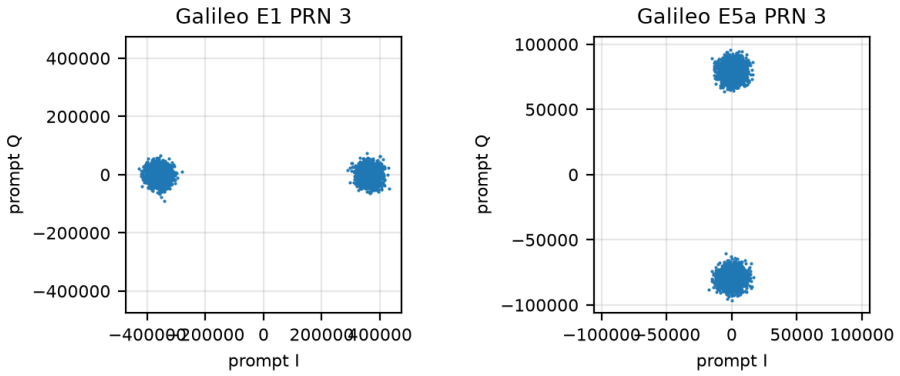


Figure 15: Tracking: prompt correlator outputs of the strongest satellite per band. Two tight clusters on one axis (the ±1 data bits; GNSS-SDR's L5 tracking puts the data component on the Q axis), or a single one for a dataless pilot (Galileo E1-C and E5a-Q, tracked here), mean the PLL is phase-locked; a ring would mean it is not. `Prompt_I`/`Prompt_Q` are 32-bit floats (sums of the samples over one integration period, in file units): they never clip, only overflowing beyond $\pm 3.4 \times 10^{38}$, far above the $N \cdot \text{full scale} \cdot \sqrt{2}$ a sum of N samples can reach.

Band	Ch.	PRN	Median C/N_0 [dB-Hz]	Phase-locked [%]	Losses of lock	Tracked [s]
Galileo E1	0	34	46.1	98	0	299
Galileo E1	1	3	46.6	87	2	298
Galileo E1	2	30	40.9	99	0	299
Galileo E1	3	8	44.4	81	2	298
Galileo E1	4	32	45.7	68	7	299
Galileo E1	5	5	44.8	98	0	298
Galileo E1	6	14	44.7	98	0	300
Galileo E1	7	15	44.5	43	8	300
Galileo E5a	8	14	48.9	31	1	275
Galileo E5a	9	34	48.9	30	1	274
Galileo E5a	10	30	43.5	52	0	272
Galileo E5a	11	5	46.8	53	0	274
Galileo E5a	12	3	51.6	49	0	244
Galileo E5a	13	8	45.9	48	0	235
Galileo E5a	14	32	50.4	37	0	144
Galileo E5a	15	15	49.0	30	0	114

Table 15: Tracking: per channel and satellite (phase-locked: carrier lock test > 0.8).

6.1.11 Telemetry decoder (Galileo E1 + E5a)

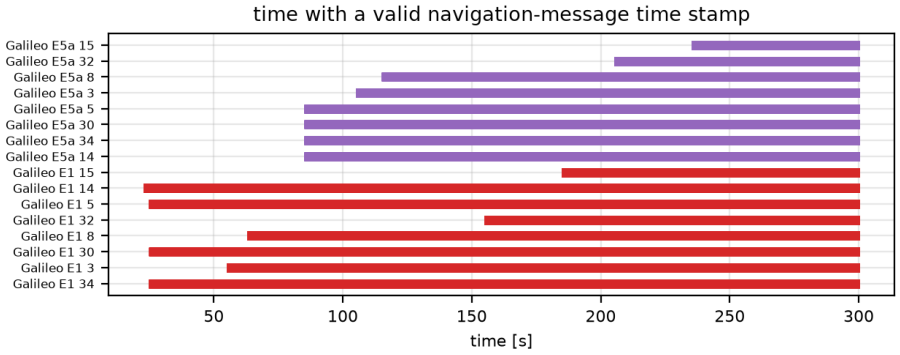


Figure 16: Telemetry decoder: per band and satellite, when the decoder delivered a valid transmit time (the first one comes after the first complete frame/message).

Band	PRN	Messages decoded	Ephemeris	First valid at [s]	TOW [s]
Galileo E1	3	24	yes	55.2	414080.0
Galileo E1	5	26	yes	25.2	414050.0
Galileo E1	8	23	yes	63.2	414088.0
Galileo E1	14	26	yes	23.2	414048.0
Galileo E1	15	10	yes	185.2	414210.0
Galileo E1	30	26	yes	25.2	414050.0
Galileo E1	32	13	yes	155.2	414180.0
Galileo E1	34	26	yes	25.2	414050.0
Galileo E5a	3	11	yes	105.4	414130.3
Galileo E5a	5	12	yes	85.4	414110.3
Galileo E5a	8	10	yes	115.4	414140.3
Galileo E5a	14	11	yes	85.4	414110.3
Galileo E5a	15	3	yes	235.4	414260.3
Galileo E5a	30	12	yes	85.4	414110.3
Galileo E5a	32	5	yes	205.4	414230.3
Galileo E5a	34	12	yes	85.4	414110.3

Table 16: Telemetry decoder: Galileo E1 I/NAV words, Galileo E5a F/NAV pages decoded, ephemeris obtained, and the first valid time of week.

6.1.12 Observables (Galileo E1 + E5a)

Pseudorange, carrier phase and Doppler of every channel at common receiver epochs.

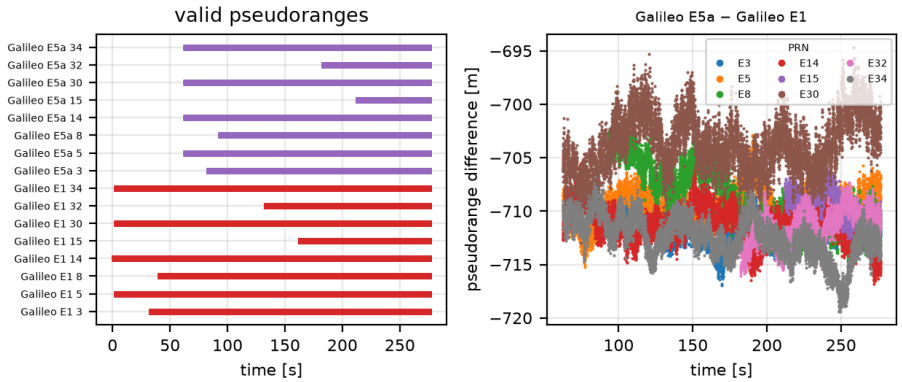


Figure 17: Observables: epochs with a valid pseudorange per band and satellite, and the difference between the two bands' pseudoranges of the same satellite (a constant receiver inter-band bias, the same for every satellite, plus the slowly varying ionospheric delay).

Difference	PRN	Common epochs	of run [%]	Mean [m]	Std [m]	Drift [m/s]
Galileo E5a − Galileo E1	3	9563	64	-711.43	1.45	-0.0075
Galileo E5a − Galileo E1	5	10437	70	-709.61	1.55	0.0065
Galileo E5a − Galileo E1	8	9128	61	-708.81	2.89	-0.0451
Galileo E5a − Galileo E1	14	10437	70	-711.16	1.64	-0.0082
Galileo E5a − Galileo E1	15	3147	21	-709.62	1.46	-0.0504
Galileo E5a − Galileo E1	30	10457	70	-703.72	2.59	0.0079
Galileo E5a − Galileo E1	32	4647	31	-711.39	1.51	0.0347
Galileo E5a − Galileo E1	34	10437	70	-712.45	1.88	-0.0177

Table 17: Observables: pseudorange difference between the two bands of each satellite at common epochs (similar means = shared receiver bias; std = code noise of both bands).

6.1.13 PVT (Galileo E1 + E5a)

Position from the pseudoranges and the decoded ephemeris, compared with the surveyed antenna position (-23.209433°, -45.874875°, 596.42 m).

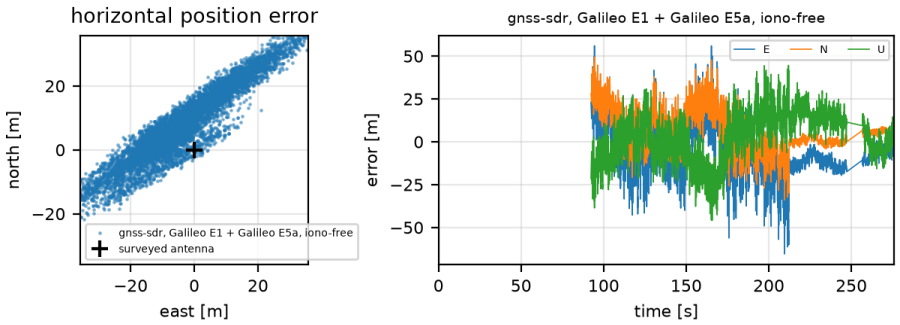


Figure 18: PVT: position error with respect to the surveyed antenna position. gnss-sdr: GNSS-SDR's own RTKLIB solution.

Solution	**gnss-sdr, Galileo E1 + Galileo E5a, iono-free**
Solutions	8642
Run with fix [%]	64
Mean E / N / U [m]	-4.8 / 6.7 / 2.8
Std E / N / U [m]	15.5 / 11.9 / 14.2
RMS 2-D [m]	21.2
RMS 3-D [m]	25.6
95% horiz. [m]	40.3

Table 18: PVT: position error statistics with respect to the surveyed antenna (mean = accuracy, std = precision; % of the run's seconds with a position; gnss-sdr outputs several solutions per second, gnss_spp.py computes one per second).

6.2 Second record

Parameter	Channel 0	Channel 1
Start time	2026-10-22 19:30:07 UTC	
Frequency	1575.420 MHz (L1)	1227.600 MHz (L2)
Duration	300.0 s	300.0 s
Sample rate	4.000 MS/s	4.000 MS/s
Sample type	signed complex 16-bit samples (I+Q), little-endian, as received from the X310 (sc16)	
Gain	25.0 dB	25.0 dB
Antenna	RX1	RX2
File size	4800.0 MB	4800.0 MB
Samples written	1,200,000,000	1,200,000,000
Lost samples	0	0
Clock source	external	
Time source	internal	
Motherboard	USRP X310	
Daughterboard	TwinRX RX0	TwinRX RX1
Recorder	UHD Python API	
GNSS-SDR configuration	`l1ca16bits4msps_l2c16bits4msps.conf`	

Table 19: Metadata for recording `x310_20261022T193000Z`, captured starting 2026-10-22T19:30:07.040026Z.

6.2.1 Signal source

Raw I/Q samples of each channel analysed within the window [10, 10.5] s after the start recording instant.

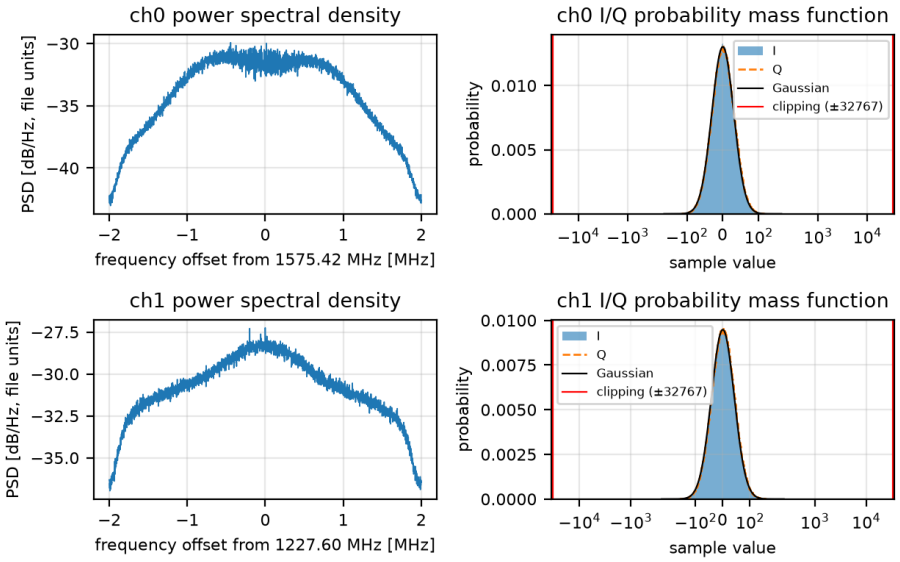


Figure 19: Signal source: power spectral density (the receiver passband over a flat thermal-noise floor; GNSS signals are below the noise, so only interference would stand out) and probability mass functions of the integer I/Q sample values (Gaussian noise, centred, not clipped).

Channel	**ch0**	**ch1**
Frequency [MHz]	1575.42	1227.60
Rate [MS/s]	4	4
Format	ci16_le	ci16_le
Bytes per complex sample	4 (2 for I, 2 for Q)	4 (2 for I, 2 for Q)
Full scale	±32767	±32767
Gain [dB]	25	25
Std I / Q	30.2 / 31.1	42.1 / 42.1
Mean I / Q	-0.04 / -0.13	0.28 / 0.05
Min / max I	-151 / 154	-227 / 208
Min / max Q	-150 / 164	-208 / 210
Headroom (full scale / std)	1069.3	778.5
Clipped [%]	0.000	0.000
Lost samples	0	0

Table 20: Signal source: sample statistics per channel (headroom: how many standard deviations of the noise fit before clipping).

6.2.2 Signal conditioner

Channel	ch0 (GPS L1 C/A, Galileo E1)	ch1 (GPS L2C)
File format	ci16_le	ci16_le
Source item type	ishort	ishort
Data type adapter	Ishort_To_Complex	Ishort_To_Complex
Input filter	Pass_Through	Pass_Through
Resampling	Pass_Through	Pass_Through
Rate [MS/s]	4 → 4	4 → 4

Table 21: Signal conditioner: recorded format and rate to gnss-sdr's internal `gr_complex` stream; both bands are read in lockstep by one `Multichannel_File_Signal_Source`.

6.2.3 Acquisition

Two-dimensional search over code delay and Doppler (one channel per band dumped; the others behave alike).

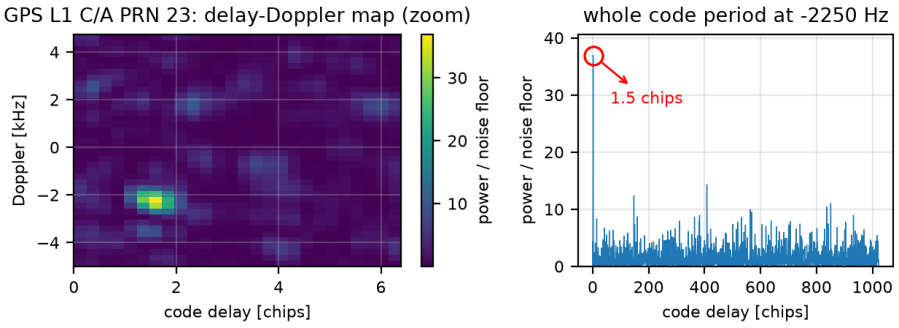


Figure 20: Acquisition, GPS L1 C/A: delay–Doppler search of PRN 23; one sharp peak (test statistic 52 vs threshold 44) at -2250 Hz and 1.5 chips.

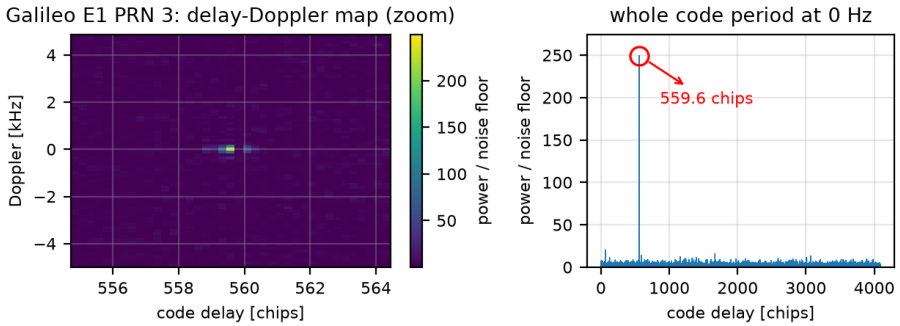


Figure 21: Acquisition, Galileo E1: delay–Doppler search of PRN 3; one sharp peak (test statistic 338 vs threshold 48) at 0 Hz and 559.6 chips.

Band	Attempts	Detections	Detected PRNs	Best statistic / threshold
GPS L1 C/A	3	1	23	1.2
GPS L2C	not dumped (search grid too large); acquired satellites: see the tracking table			
Galileo E1	6	1	3	7.0

Table 22: Acquisition: search attempts of the dumped channel. Attempts on satellites that are not in view (or do not broadcast that signal, e.g. older GPS satellites on L5/L2C) are rejected below the threshold.

6.2.4 Tracking

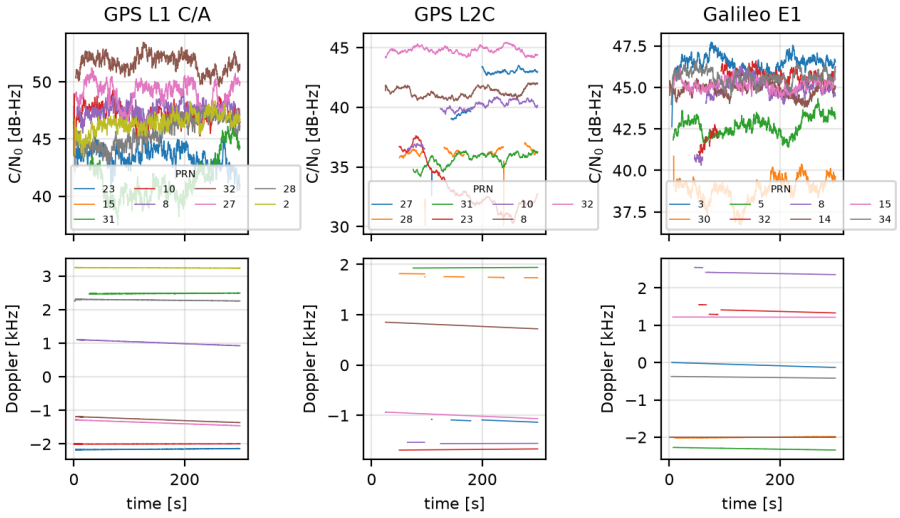


Figure 22: Tracking: carrier-to-noise density ratio and carrier Doppler per satellite, while phase-locked (smooth Doppler curves = loops following the real satellite motion).

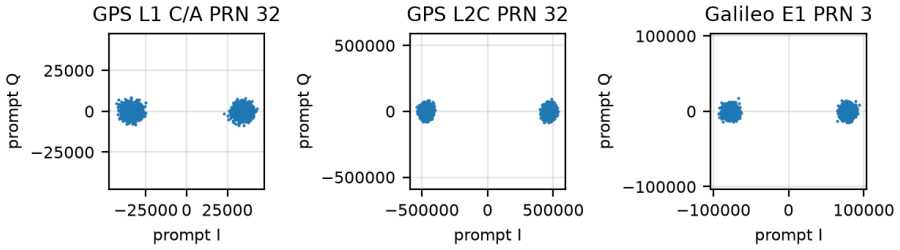


Figure 23: Tracking: prompt correlator outputs of the strongest satellite per band. Two tight clusters on one axis (the ±1 data bits; GNSS-SDR's L5 tracking puts the data component on the Q axis), or a single one for a dataless pilot (Galileo E1-C and E5a-Q, tracked here), mean the PLL is phase-locked; a ring would mean it is not. `Prompt_I`/`Prompt_Q` are 32-bit floats (sums of the samples over one integration period, in file units): they never clip, only overflowing beyond $\pm 3.4 \times 10^{38}$, far above the $N \cdot \text{full scale} \cdot \sqrt{2}$ a sum of N samples can reach.

Band	Ch.	PRN	Median C/N_0 [dB-Hz]	Phase-locked [%]	Losses of lock	Tracked [s]
GPS L1 C/A	0	23	43.6	94	0	298
GPS L1 C/A	1	15	28.7	0	1	16
GPS L1 C/A	1	31	42.2	95	1	273
GPS L1 C/A	2	10	46.9	100	0	300
GPS L1 C/A	3	8	47.4	92	0	297
GPS L1 C/A	4	32	51.4	95	0	298
GPS L1 C/A	5	27	48.7	96	0	298
GPS L1 C/A	6	28	45.0	96	0	298
GPS L1 C/A	7	2	45.8	100	0	298
GPS L2C	8	27	39.5	50	3	275
GPS L2C	9	28	36.3	49	3	275
GPS L2C	10	31	35.8	90	0	251
GPS L2C	12	23	32.7	91	0	276
GPS L2C	13	10	40.0	75	1	276
GPS L2C	14	8	41.1	100	0	275
GPS L2C	15	32	44.8	100	0	275
Galileo E1	16	3	46.5	100	0	297
Galileo E1	17	30	38.9	99	0	292
Galileo E1	18	5	42.7	99	0	297
Galileo E1	19	32	45.5	80	4	293
Galileo E1	20	8	44.8	84	3	297
Galileo E1	21	14	44.9	100	0	299
Galileo E1	22	15	45.1	98	0	299
Galileo E1	23	34	45.6	100	0	297

Table 23: Tracking: per channel and satellite (phase-locked: carrier lock test > 0.8).

6.2.5 Telemetry decoder

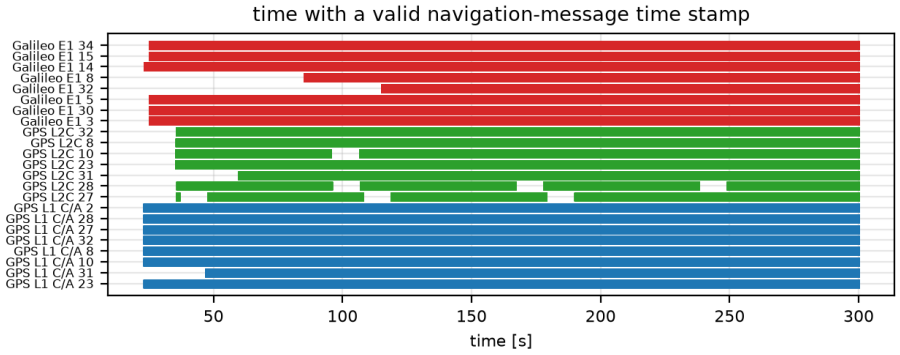


Figure 24: Telemetry decoder: per band and satellite, when the decoder delivered a valid transmit time (the first one comes after the first complete frame/message).

Band	PRN	Messages decoded	Ephemeris	First valid at [s]	TOW [s]
GPS L1 C/A	2	42	yes	23.0	415848.0
GPS L1 C/A	8	47	yes	23.0	415848.0
GPS L1 C/A	10	43	yes	23.0	415848.0
GPS L1 C/A	23	47	yes	23.0	415848.0
GPS L1 C/A	27	47	yes	23.0	415848.0
GPS L1 C/A	28	47	yes	23.0	415848.0
GPS L1 C/A	31	43	yes	47.0	415872.0
GPS L1 C/A	32	47	yes	23.0	415848.0
GPS L2C	8	13	yes	35.5	0.0
GPS L2C	10	9	yes	35.4	0.0
GPS L2C	23	15	yes	35.4	0.0
GPS L2C	27	5	yes	35.7	0.0
GPS L2C	28	–	no	35.8	0.0
GPS L2C	31	12	yes	59.8	0.0
GPS L2C	32	13	yes	35.6	0.0
Galileo E1	3	27	yes	25.1	415850.0
Galileo E1	5	26	yes	25.1	415850.0
Galileo E1	8	21	yes	85.1	415910.0
Galileo E1	14	26	yes	23.1	415848.0
Galileo E1	15	27	yes	25.1	415850.0
Galileo E1	30	27	yes	25.1	415850.0
Galileo E1	32	18	yes	115.1	415940.0
Galileo E1	34	27	yes	25.1	415850.0

Table 24: Telemetry decoder: GPS L1 C/A LNAV subframes, GPS L2C CNAV messages, Galileo E1 I/NAV words decoded, ephemeris obtained, and the first valid time of week.

6.2.6 Observables

Pseudorange, carrier phase and Doppler of every channel at common receiver epochs.

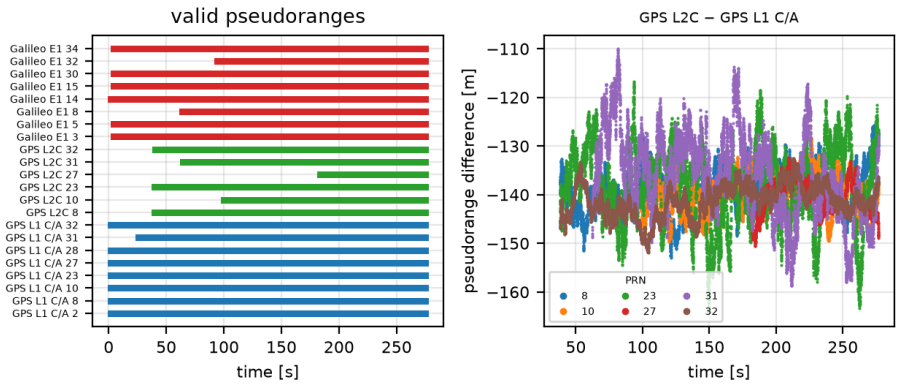


Figure 25: Observables: epochs with a valid pseudorange per band and satellite, and the difference between the two bands' pseudoranges of the same satellite (a constant receiver inter-band bias, the same for every satellite, plus the slowly varying ionospheric delay).

Difference	PRN	Common epochs	of run [%]	Mean [m]	Std [m]	Drift [m/s]
GPS L2C – GPS L1 C/A	8	11932	80	-138.80	4.22	0.0218
GPS L2C – GPS L1 C/A	10	8950	60	-139.45	3.99	0.0142
GPS L2C – GPS L1 C/A	23	11937	80	-138.23	7.93	-0.0048
GPS L2C – GPS L1 C/A	27	4756	32	-140.52	3.36	-0.0064
GPS L2C – GPS L1 C/A	31	10717	72	-135.85	8.56	-0.0721
GPS L2C – GPS L1 C/A	32	11928	80	-141.29	3.29	0.0180

Table 25: Observables: pseudorange difference between the two bands of each satellite at common epochs (similar means = shared receiver bias; std = code noise of both bands).

6.2.7 PVT

Position from the pseudoranges and the decoded ephemeris, compared with the surveyed antenna position (-23.209433°, -45.874875°, 596.42 m).

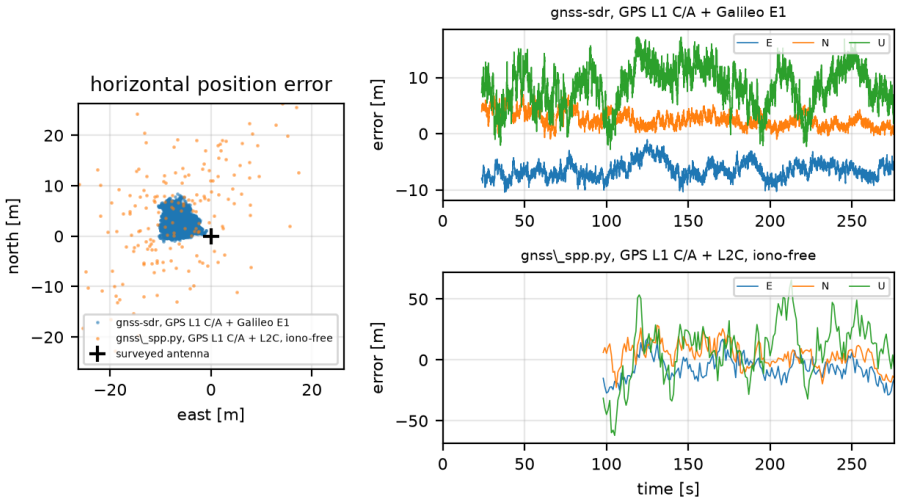


Figure 26: PVT: position error with respect to the surveyed antenna position. gnss-sdr: GNSS-SDR's own RTKLIB solution (single frequency, broadcast ionosphere model). gnss_spp.py: computed from gnss-sdr's observables and decoded GPS ephemeris, removing the ionospheric delay with the second band (pseudoranges carrier-smoothed over 100 s).

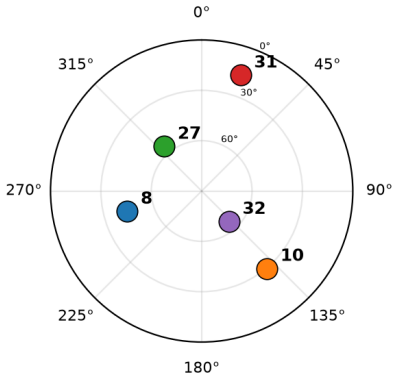


Figure 27: PVT: sky plot of the GPS satellites in the gnss_spp.py dual-frequency solution (azimuth, elevation) during the recording.

Solution	**gnss-sdr, GPS L1 C/A + Galileo E1**	**gnss_spp.py, GPS L1 C/A + L2C, iono-free**
Solutions	13139	179
Run with fix [%]	92	65
Mean E / N / U [m]	-6.5 / 2.6 / 8.4	-6.9 / 4.1 / 8.3
Std E / N / U [m]	1.5 / 1.4 / 3.7	9.6 / 10.1 / 23.8
RMS 2-D [m]	7.3	16.1
RMS 3-D [m]	11.7	29.9
95% horiz. [m]	9.5	28.1

Table 26: PVT: position error statistics with respect to the surveyed antenna (mean = accuracy, std = precision; % of the run's seconds with a position; gnss-sdr outputs several solutions per second, gnss_spp.py computes one per second).